
FJM30 Wideband Integrated RF Processing Device | Product Datasheet

The FJM30 is a highly integrated wideband RF processing device that combines dual-channel direct-conversion RF transceiver functionality with a heterogeneous computing architecture consisting of programmable hardware acceleration and embedded multi-core processing.

The architecture enables deterministic low-latency signal processing, adaptive waveform control, and real-time spectrum analysis while minimizing external system complexity.

1. Heterogeneous Processing Architecture

FJM30 integrates a tightly coupled programmable hardware fabric and embedded processing subsystem designed for hardware/software co-optimization.

ARM Processing Subsystem

- Dual-core ARM architecture processors
- Maximum operating frequency: up to 866 MHz
- 32-bit processing architecture
- Integrated L1 cache architecture
- High-bandwidth AXI-based interconnect
- Hardware interrupt controller
- Multi-master bus arbitration

Programmable Logic Fabric

- 85K equivalent logic capacity
- 200 hardware multiplier/accumulator units

Memory Subsystem

- External DDR3/DDR3L/LPDDR2 support
- 16-bit interface
- Up to 1066 MT/s
- On-Chip Memory: 5 Mb of high-speed memory plus >100 Kb of distributed logic memory. plus 256 Mb SPI Flash for autonomous boot

2. RF Subsystem Specifications

The RF subsystem integrates a wideband dual-channel direct-conversion transceiver (2T2R) optimized for high dynamic range and low phase noise operation.

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- Frequency Coverage: 50 MHz – 6.0 GHz
 - Channel Programmable Bandwidth: 200 kHz – 56 MHz
 - Aggregate usable bandwidth: 100 MHz

Receiver Performance (Typical)

- ADC Resolution: 12-bit
- Noise Figure: 2.4 dB @1.5 GHz, < 5.6 dB across band
- IIP3: -36 dBm @1.5 GHz
- SFDR: 55 dBc
- RSSI with gain compensation

Transmitter Performance (Typical)

- DAC Resolution: 12-bit
- Maximum Conducted Output Power: +5 dBm (typical)
- TX EVM: < 1% @1.5 GHz, < 2% across band
- OIP3: up to 23 dBm
- ACLR: 58 dB (64QAM)
- Carrier Leakage: 57 dBc
- TX Noise Floor: down to -158 dBm/Hz

Frequency Synthesizer

- Fractional-N PLL architecture
- Frequency resolution: 2.4 Hz
- Switching time: ~80 μ s
- Phase Noise (Typical): -108 dBc/Hz @10 kHz offset (1.5 GHz), -123 dBc/Hz @1 MHz offset
- Hopping rate: up to 10,000 hops/s
- Supports TDD and FDD

3. Digital Signal Processing & Logic

The programmable fabric supports deep pipelined signal processing architectures enabling high-throughput baseband processing such as:

- Digital down-conversion (DDC)
- Digital up-conversion (DUC)

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- FIR/CIC filtering
 - FFT-based spectral analysis
 - Custom protocol acceleration
 - Waveform Generation: Built-in support for AM/FM, PSK/QAM, and custom profiles.
 - Correction: Digital IQ imbalance and DC offset correction.

4. Connectivity & Interfaces

A comprehensive suite of industry-standard interfaces ensures seamless system integration.

- Network/Data: Gigabit Ethernet, USB 2.0 OTG, SD/SDIO.
- Control/Bus: CAN, UART, SPI, I²C, and user-accessible GPIO.

5. Power Supply

- 1 V (1.5 A max.) ; 1.8 V (2 A max.) ; 3.3 V (1 A max.) ;

6. Physical & Development

- Package: 625-ball BGA, 17 mm x 17 mm x 1.66 mm, 0.65 mm pitch.
- Software Support: Compatible with Embedded Linux, RTOS, and Bare-metal environments.
- Development Flow: The device supports industry-standard FPGA design toolchains for RTL-based hardware development (Verilog/VHDL), hardware debugging via JTAG, integrated logic analysis, and remote firmware upgrade mechanisms.

Applications

- MIMO multi-polarization transceiver communication.
- Software-defined radio (SDR) platforms.
- Multi-channel Digital/Analog signal sources.
- Real-time spectrum analyzers and frequency counters.

Block Diagram

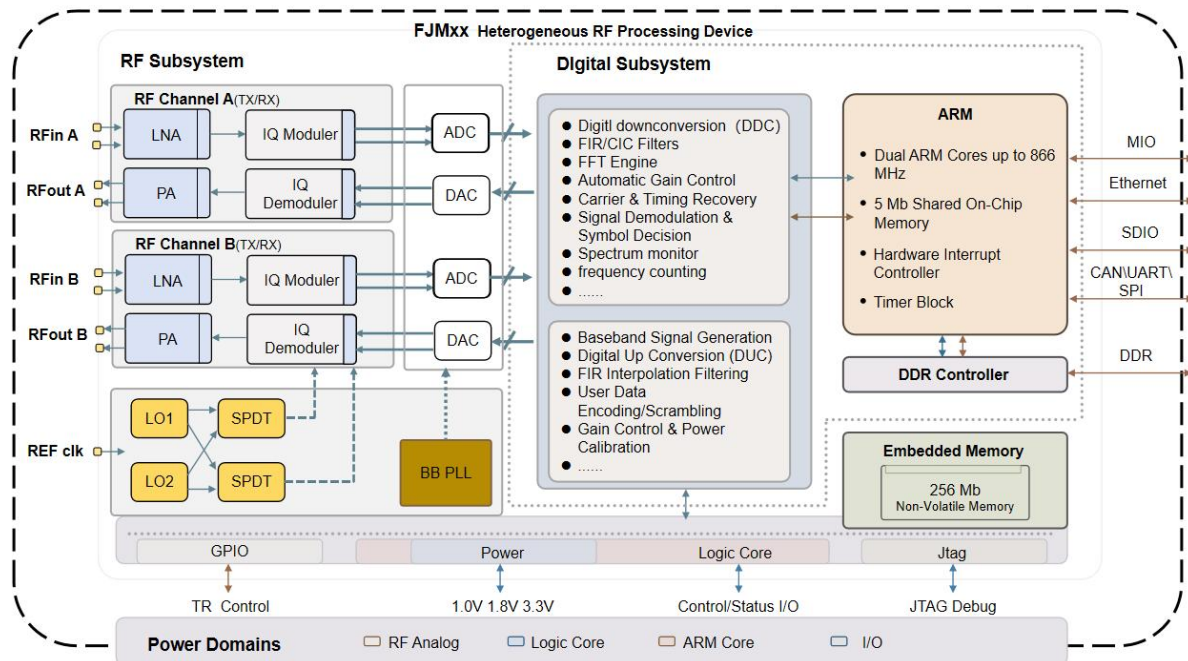
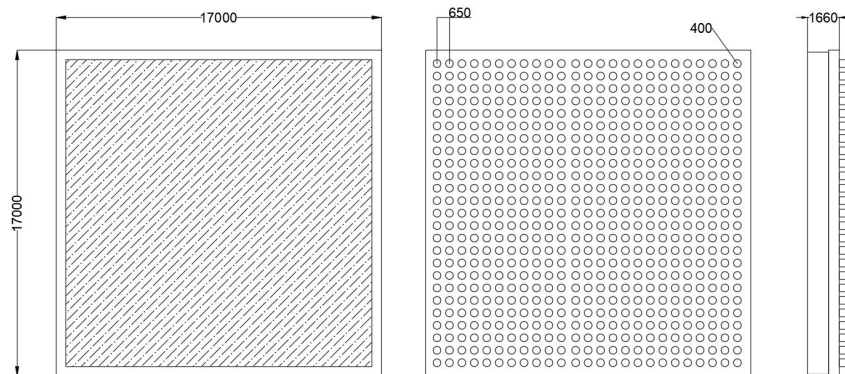


Figure 1: Block Diagram

Package Information



- Package Type: BGA
- Package Dimensions: 17.0×17.0×1.66mm;
- Ball Count: 625 balls;
- pitch: 0.65mm